Real Keynesian Models and Sticky Prices

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- In many macro models, the key element that allows for demand shocks (optimism, positive sentiment, good news, possibly lax credit,...) to have expansionary effects is the presence of sticky prices.
- There is an alternative (much smaller) literature which suggest that sticky prices may not be necessary for demand shocks to be expansionary. (Ex: Angeletos-La'O, Angeletos-Lian, Angeletos-Collard-Dellas, Guerrieri-Lorenzoni, Lorenzoni, Beaudry-Portier, Beaudry-Galizia-Portier,... etc $\rightsquigarrow$ *Real Keynesian* models
- Question addressed in this paper: should we care?

Introduction: The Question

- Suppose one accepts the evidence that nominal prices are sticky in a New Keynesian way (meaning that it implies demand non-neutrality)
- Is it important to know whether demand shocks would be expansionary even in their absence?
- In particular, is it important for
  1. our understanding of how shocks affect the economy?
  2. our understanding the conduct of monetary policy?
- It is.

Introduction: Two Contributions

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  1. Propose a new class of simple extensions of the *New Keynesian* model (a *Real Keynesian* model) that has very different implications for monetary policy **when prices are sticky**.
  2. Propose a novel way to identify shocks in SVARs
     (I will not go too much into details on this contribution)

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Roadmap

1. Theory & Estimation
2. Dissecting the Results Using a New SVAR Approach
3. Zero Lower Bound and Missing Deflation

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The Real Structure of the Simplest NK Model

- No technology shock $c_t = \ell_t$
- Model with sticky prices:

$$\ell_t = E_t \ell_{t+1} - \alpha_r (i_t - E_t \pi_{t+1}) + d_t \qquad \text{Euler Equation (EE)}$$
$$\pi_t = \beta E_t \pi_{t+1} + \kappa \; mc_t \qquad \text{Phillips Curve (PC)}$$

- Marginal cost is assumed to depend on labor market tightness (real wage) $\rightsquigarrow$ $mc_t = \gamma_\ell \ell_t$
- When prices are fully flexible:

$$\ell_t = E_t \ell_{t+1} - \alpha_r r_t + d_t \qquad \text{Euler Equation (EE)}$$
$$mc_t = 0 = \gamma_\ell \ell_t \qquad \text{Aggregate Supply (AS)}$$

The Real Structure of the Simplest NK Model

Flex price NK model :

$$\ell_t = E_t \ell_{t+1} - \alpha_r r_t + d_t \quad \text{(EE)}$$
$$0 = \gamma_\ell \ell_t \quad \text{(AS)}$$

The Real Structure of the Simplest NK Model

i.i.d. case :

$$\begin{aligned} \ell_t &= \qquad\quad - \alpha_r r_t + d_t \quad \text{(EE)} \\ 0 &= \gamma_\ell \ell_t \qquad\qquad\qquad\quad \text{(AS)} \end{aligned}$$

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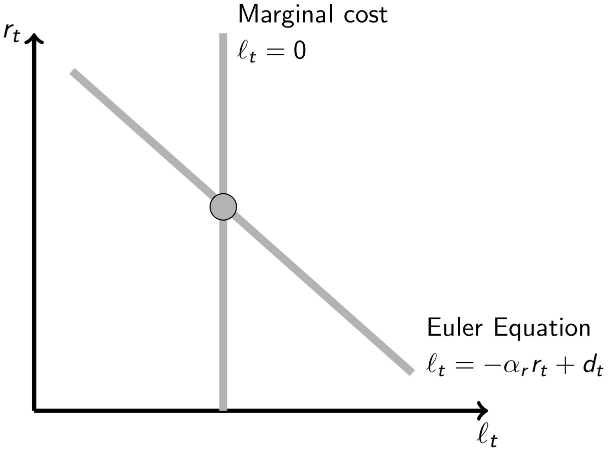

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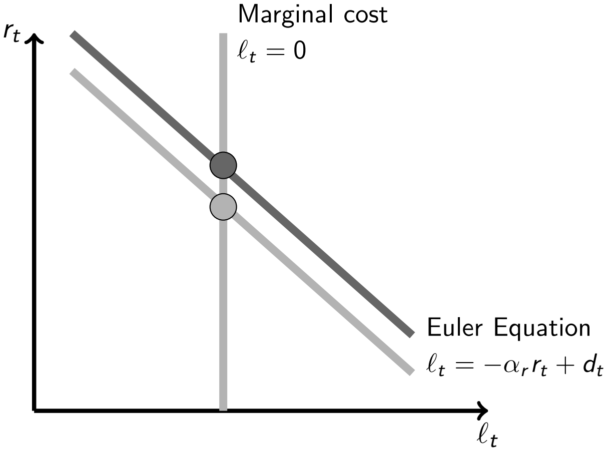

The Real structure of the Simplest NK Model

- Let's have a more general model in which AS is not infinitely sloped.
- Assume now that marginal cost also depend on the real interest rate r (*cost channel*)

$$mc_t = \gamma_\ell \ell_t + \gamma_r r_t$$

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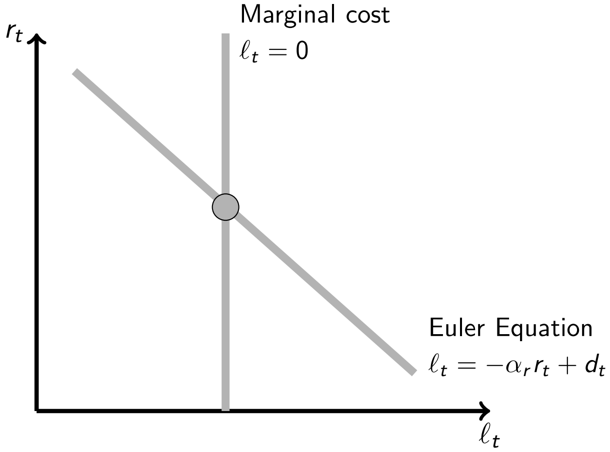

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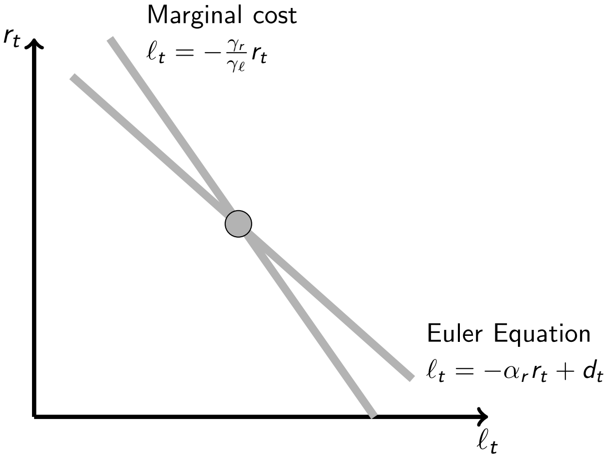
- Assume γ_r is small (compared to γ_ℓ)



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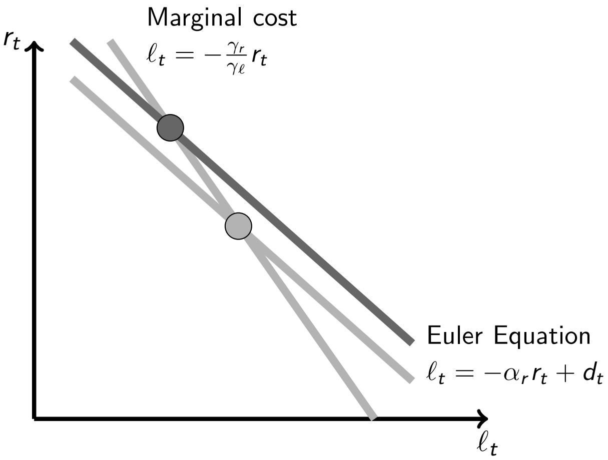

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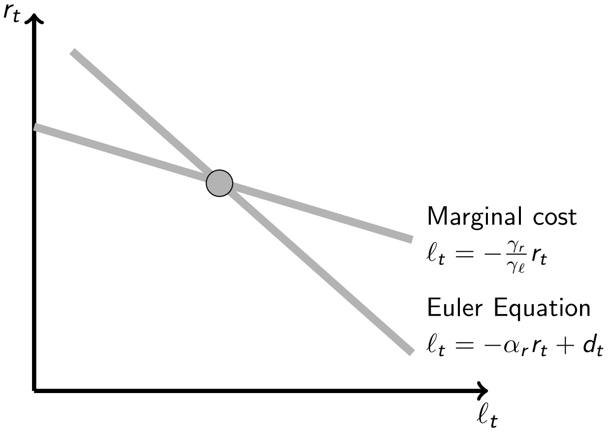

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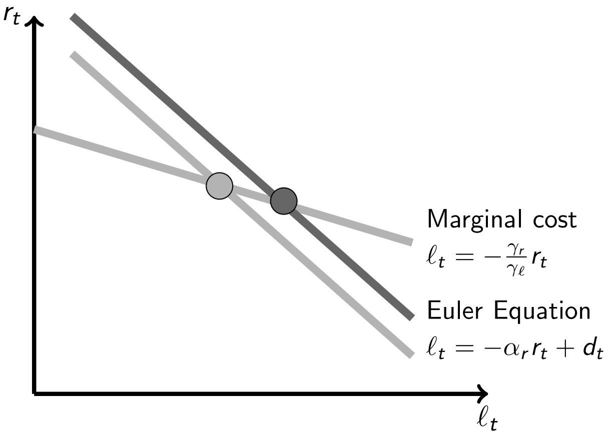

Towards An Extended Model

- Importance of the cost channel: $\frac{\gamma_r}{\gamma_\ell} \lessgtr \alpha_r$
- In the *i.i.d.* case, the model is *Real Keynesian* if $\frac{\gamma_r}{\gamma_\ell} > \alpha_r$
- Need to go beyond the *i.i.d.* case
- $\rightsquigarrow$ Expectations in the Euler equation will matter

Extended Sticky Price Linearized Model

$$\ell_t = \alpha_\ell E_t \ell_{t+1} - \alpha_r(i_t - E_t\pi_{t+1}) + d_t \qquad \text{Euler Equation (EE)}$$

$$\pi_t = \beta E_t \pi_{t+1} + \kappa\big(\gamma_\ell \ell_t + \gamma_r(i_t - E_t\pi_{t+1})\big) \qquad \text{Phillips Curve (PC)}$$

- Two changes:
  - × $\alpha_\ell \leq 1$: Add asymmetric information: some households always repay their debt, some do only if it is in their interest, with psychological cost of defaulting $\rightsquigarrow$ positively sloped cost of funds $\rightsquigarrow$ *discounted EE*
  - × $\gamma_r \geq 0$: Firms need to borrow to pay for intermediate inputs before production $\rightsquigarrow$ *cost channel*
- Nothing novel, except for putting them together.
- Note: standard NK model: $\alpha_\ell = 1$, $\gamma_r = 0$
- Here only demand shock (news shock, β shock,...)
- To remember: α's for the EE, γ's for the PC

Evidence for the Cost Channel
(Preliminary)

- Estimate a Phillips Curve with cost channel (1969Q1-2006Q4)

$$\pi_t - \pi_{t-1} = a_1(\pi_{t+1} - \pi_t) + a_2(i_t - \pi_{t+1}) + a_3\ell_t + f(t) + u_t$$

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$\ell_t = UNgap$	(1)Core CPI with Trend			(2) Core CPI w/o Trend		
	OLS	OLS	IV	OLS	OLS	IV
a_1	0.49***	0.57***	0.70***	0.49***	0.56***	0.62***
	(0.087)	(0.103)	(0.087)	(0.084)	(0.093)	(0.080)
a_2		0.21**	0.31***		0.16*	0.15**
		(0.101)	(0.111)		(0.081)	(0.075)
a_3	-0.09	-0.03	0.25	-0.05	-0.07	0.11
	(0.105)	(0.119)	(0.189)	(0.095)	(0.093)	(0.214)
Observations	152	152	152	152	152	152
K-P LM Test			29.951			16.341
(idp)			(0.038)			(0.569)
J Test			12.944			16.559
(jp)			(0.740)			(0.485)
C-D Test			2.212			0.846
Time trend	Yes	Yes	Yes	No	No	No

Extended Flex Price Linearized Model

$$\ell_t = \alpha_\ell E_t \ell_{t+1} - \alpha_r r_t + d_t \qquad \text{Euler Equation (EE)}$$

$$0 = \gamma_\ell \ell_t + \gamma_r r_t \qquad \text{Marginal cost (AS)}$$

The RK condition

Result 1

With flex. prices, positive demand shocks (both current and expected future) of any persistence have a positive effect on ℓ if and only if

$$\frac{\gamma_r}{\gamma_\ell} > \frac{\alpha_r}{\qquad\qquad} \qquad (RK)$$

The RK condition

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$$\frac{\gamma_r}{\gamma_\ell} > \frac{\alpha_r}{(1-\alpha_\ell)} \qquad (RK)$$

The Model With Sticky Prices (From now on)

$$\ell_t = \alpha_\ell E_t \ell_{t+1} - \alpha_r (i_t - E_t \pi_{t+1}) + d_t \tag{EE}$$

$$\pi_t = \beta E_t \pi_{t+1} + \kappa \big(\gamma_\ell \ell_t + \gamma_r (i_t - E_t \pi_{t+1}) \big) + \mu_t \tag{PC}$$

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$$i_t = E_t \pi_{t+1} + \phi_\ell \ell_t + \nu_t \tag{Policy Rule}$$

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Result 2

For any Taylor rule $i_t = \phi_\pi \pi_t + \phi_\ell \ell_t + \widetilde{\nu}_t$ that gives determinancy, there exists a policy rule $i_t = E_t \pi_{t+1} + \phi_\ell \ell_t + \nu_t$ that produces the same allocations.

Result 3

With policy rule $\phi_\ell > 0$, the economy is determinate for all admissible parameter values.

Irrelevance Result

Result 4

With sticky prices, RK and NK configurations are not qualitatively distinguishable for demand and markup shocks.

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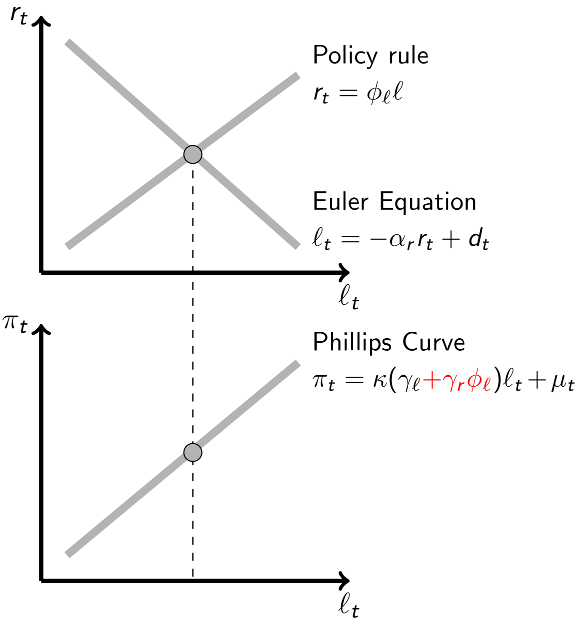

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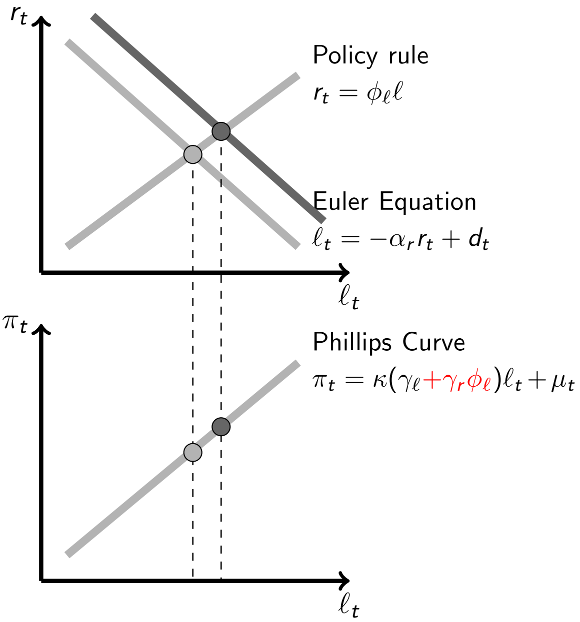

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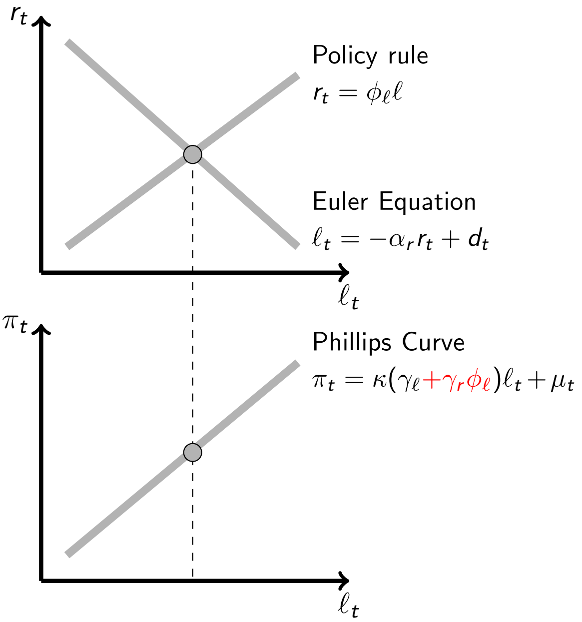

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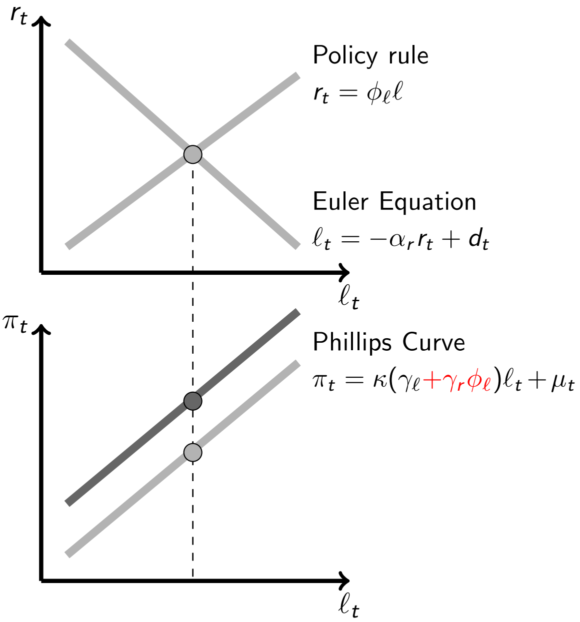

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RK Matters for Monetary Policy and Monetary Shocks

- Monetary Policy and Stabilization
- Determinacy under i peg
- Monetary Shocks

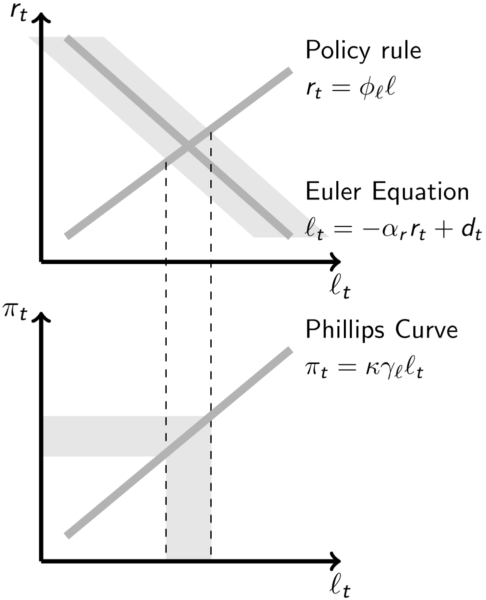
Effects of Stabilization with Demand Shocks

$$i_t = E_t \pi_{t+1} + \phi_\ell \ell_t$$

Result 5

A more aggressive policy (ϕ_ℓ larger) always decreases σ_ℓ^2 at the cost of increasing σ_π^2 iff the RK condition is satisfied.

NK Configuration ($\gamma_r = 0$, $\alpha_\ell = 1$)



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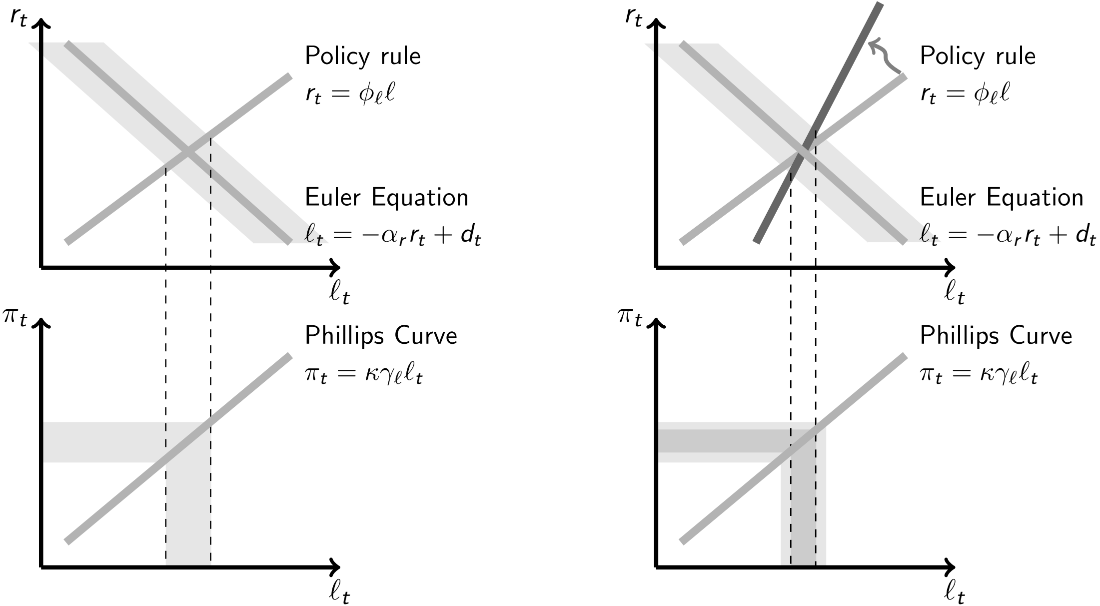
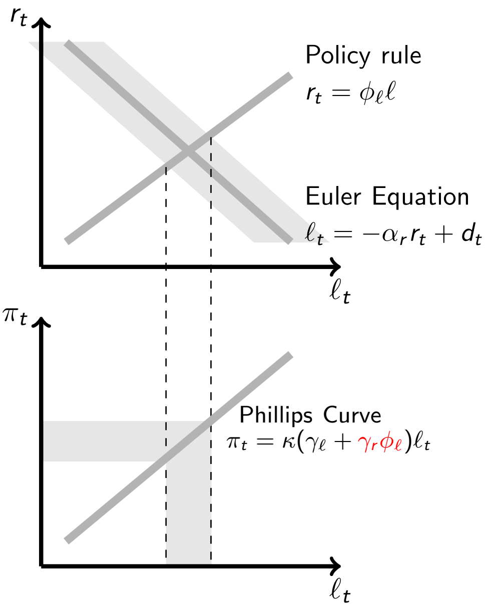


Under RK (γ_r large enough)



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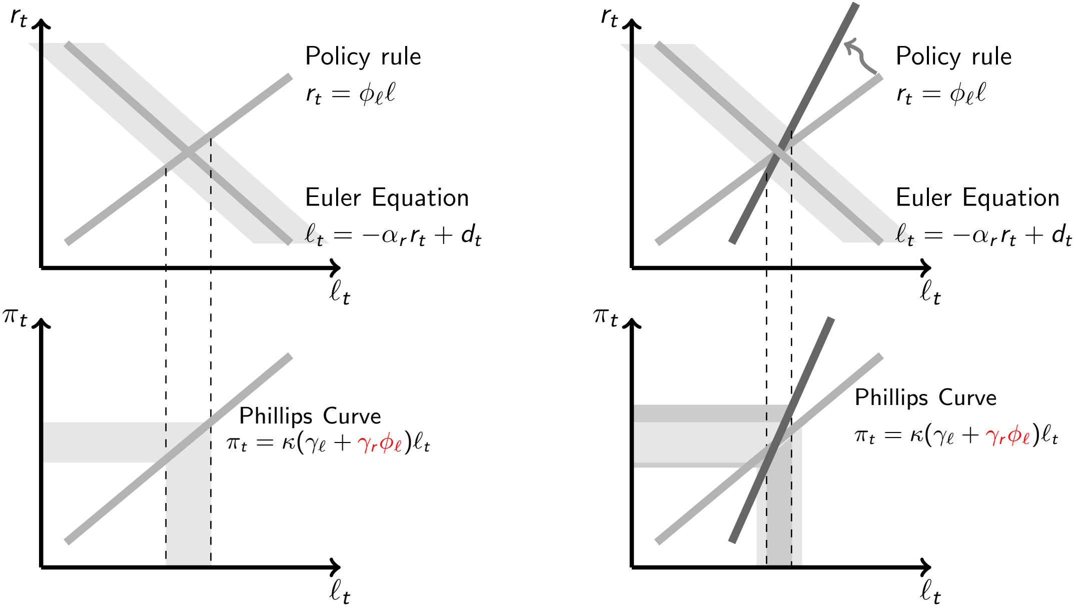

Nominal Interest Rate Peg (ZLB)

- Suppose policy goes from

$$i_t = E_t\pi_{t+1} + \phi_2\ell_t$$

to

$$i_t = 0.$$

Result 6

In the NK configuration,

× *indeterminacy*

× *in all equilibria, σ_ℓ^2 and σ_π^2 move together (conditional on demand shocks)*

In the RK configuration,

× *determinacy*

× *σ_ℓ^2 increases but σ_π^2 decreases (conditional on demand shocks)*

Monetary Shocks

Result 7

In response to a contractionary monetary shocks,

- *If the shock is not very persistent, then NK and RK cannot be distinguished.*
- *If shock is sufficiently persistent,*
  - × *it increases inflation in RK case (neo-Fisherian effect)*
  - × *it decreases inflation in the NK case*

- RK favoured if we observe both (1) persistent monetary shock that (2) do not lead to a fall in inflation
- "*Congressman Wright Patman effect*" (1970) : raising interest rates to fight inflation is like "throwing gasoline on fire"

Estimation

- Data:
  - π: GDP deflator,
  - i_t: fed funds rate,
  - ℓ_t: minus unemployment rate.
- Sample:
  - long: 1954:3- 2007:4,
  - post-Volker-deflation sample: 1983:4-2007:4
- Maximum Likelihood estimation

Result 8

Estimation shows that the model is in the Real Keynesian region.

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SVAR representation and Maximum Likelihood

- The model solution writes

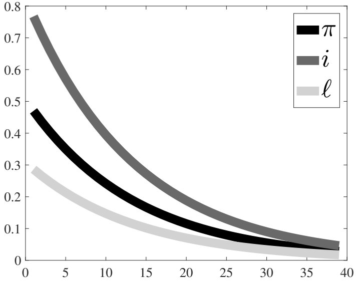
$$\underbrace{\begin{pmatrix} \pi_t \\ i_t \\ \ell_t \end{pmatrix}}_{Y_t} = A(\Theta) \begin{pmatrix} \pi_{t-1} \\ i_{t-1} \\ \ell_{t-1} \end{pmatrix} + B(\Theta) \underbrace{\begin{pmatrix} d_t \\ \mu_t \\ \nu_t \end{pmatrix}}_{S_t}$$

$$\begin{pmatrix} d_t \\ \mu_t \\ \nu_t \end{pmatrix} = \underbrace{\begin{pmatrix} \rho_d & 0 & 0 \\ 0 & \rho_\mu & 0 \\ 0 & 0 & \rho_\mu \end{pmatrix}}_{R(\Theta)} \begin{pmatrix} d_{t-1} \\ \mu_{t-1} \\ \nu_{t-1} \end{pmatrix} + \underbrace{\begin{pmatrix} \varepsilon_{dt} \\ \varepsilon_{\mu t} \\ \varepsilon_{\nu t} \end{pmatrix}}_{\varepsilon_t}$$

- This is what we use for ML in order to obtain $\widehat{\Theta}$.
- But we can also estimate in two-steps:
  - × First A, B and R (**Theorem**: identification is guaranteed if R is more sparse that triangular),
  - × Second Θ.

SVAR, Full Sample

ε_1

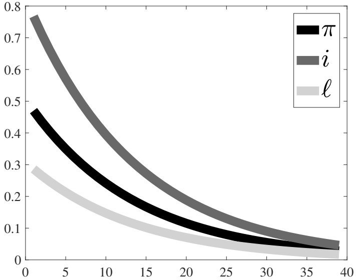


ε_2

ε_3

SVAR, Full Sample

$\varepsilon_1 \rightsquigarrow \varepsilon_d$



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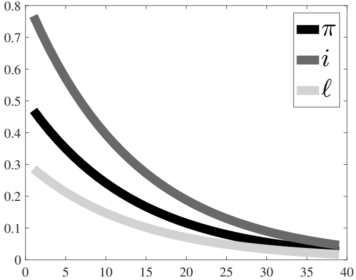

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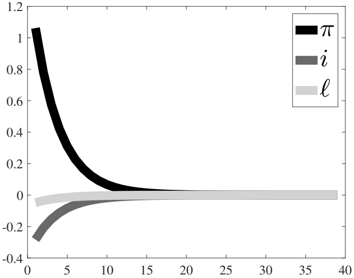

ε_3

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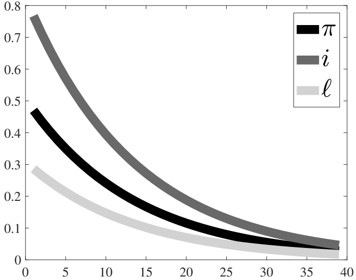
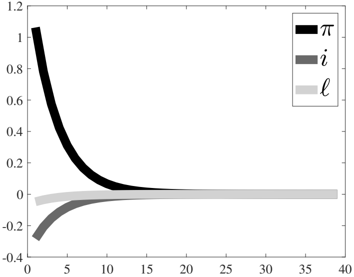
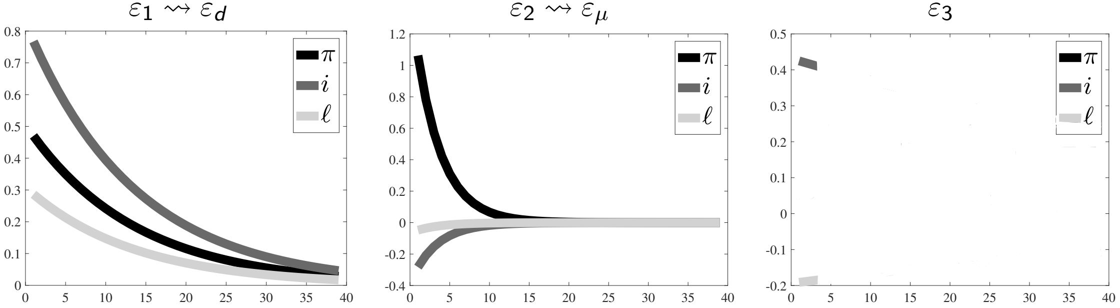


$\varepsilon_2 \rightsquigarrow \varepsilon_\mu$



ε_3

SVAR, Full Sample



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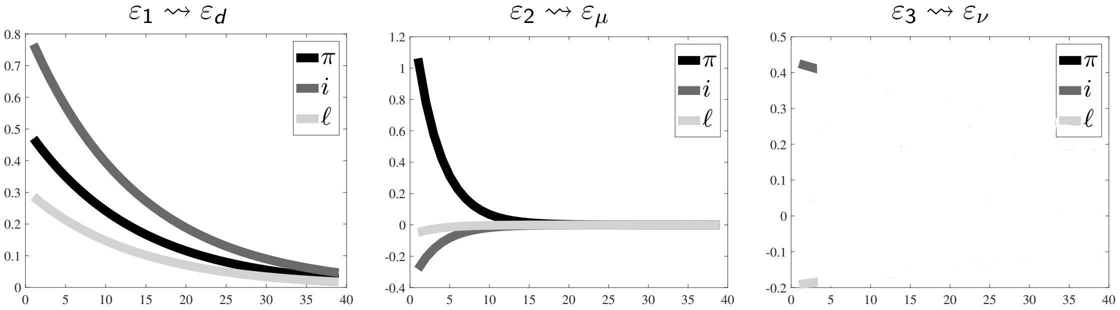

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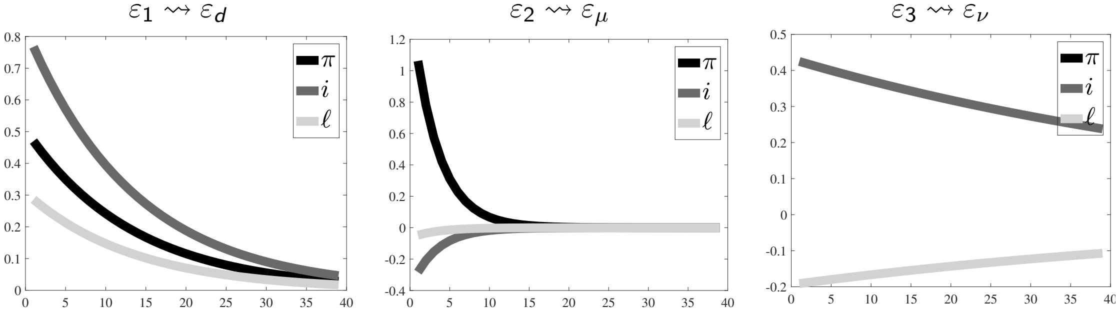

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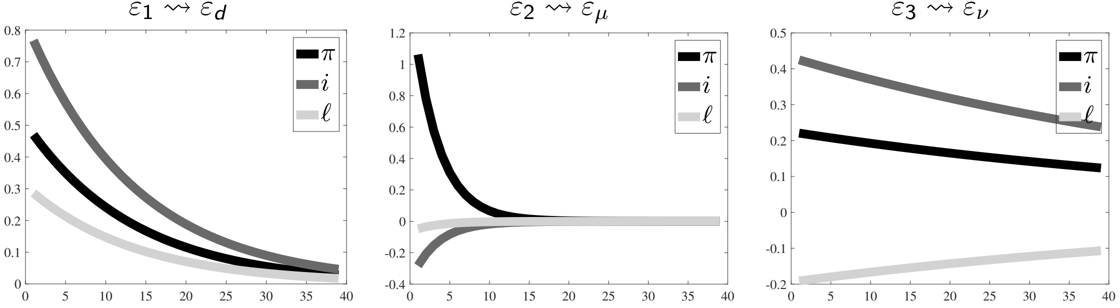
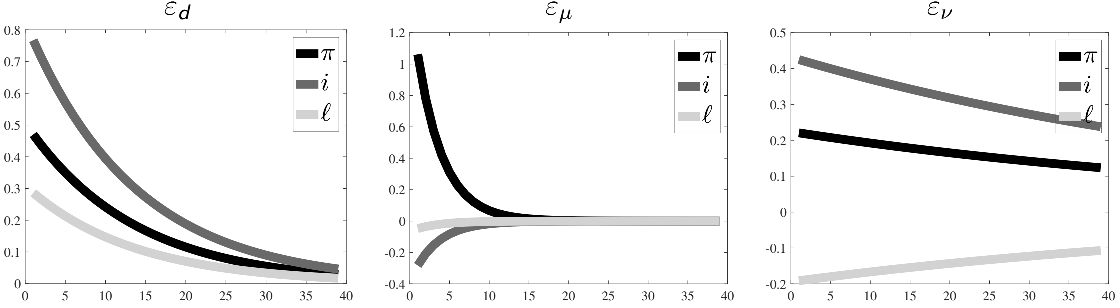


Max Likelihood Estimation, Full Sample



Three Sub-Samples

I. Pre Volker dis-inflation period (1954:3-1979:1)
II. Post Volker dis-inflation period (1983:4-2007:1)
III. Zero Lower Bound period (2009:1-2016:3)

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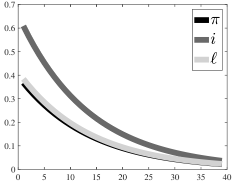
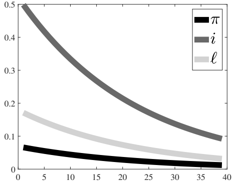
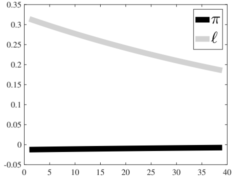
$\varepsilon_1 \rightsquigarrow \varepsilon_d$ (I)



$\varepsilon_1 \rightsquigarrow \varepsilon_d$ (II)



$\varepsilon_1 \rightsquigarrow \varepsilon_d$ (III)



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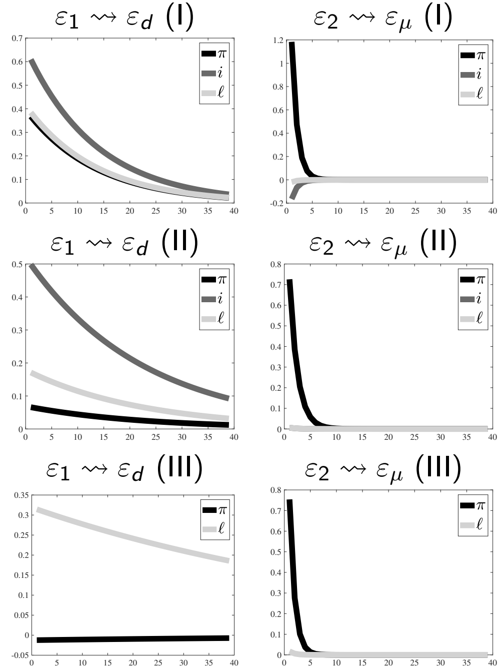

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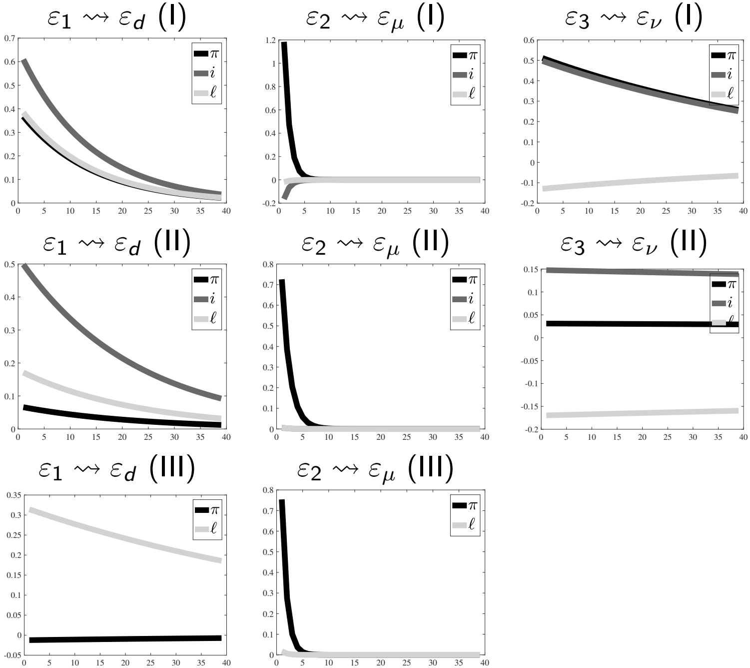


Robustness

1. Are these results robust to allowing the model to have endogenous propagation? Yes
2. Are these results robust to allowing the model to have more shocks? Yes

Can we obtain more standard effects of monetary shocks?

- Note that the RK model is not inconsistent with VAR evidence that finds temporary monetary shocks.
- But those shocks (as opposed to persistent monetary shocks) do not seem to be relevant in a simple 3-shocks (and 4-shocks) framework.

Roadmap

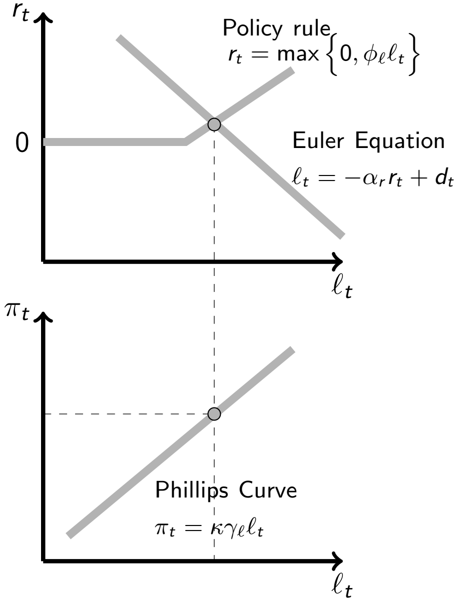
1. Theory & Estimation
2. Dissecting the Results Using a New SVAR Approach
3. Zero Lower Bound and Missing Deflation

Low Variance of Inflation at the ZLB

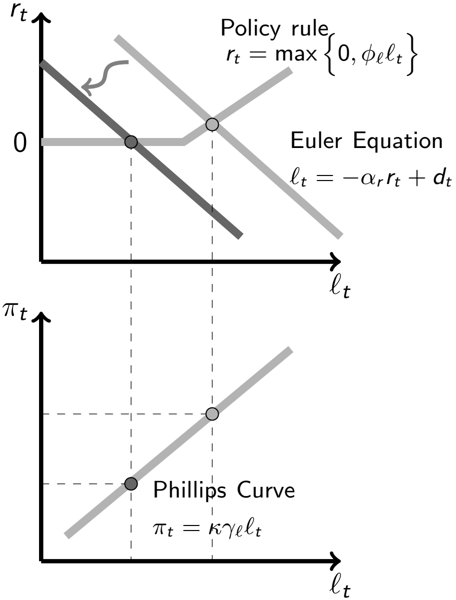
		σ_u	σ_π	σ_i
Post-Volcker	:	1.3	.9	2.5
ZLB	:	1.7	.8	.1

- Observation: the variance of inflation slightly decreased at ZLN.
- It should have increased in the NK configuration (*under the assumption that demand shocks drove the economy*)
- But this is consistent with the RK configuration

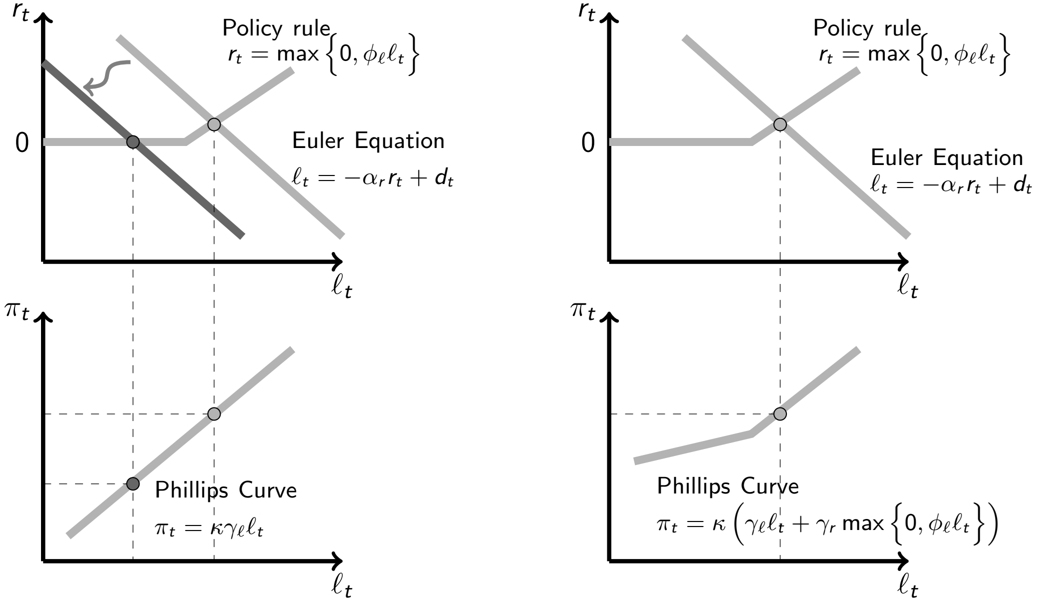
Zero Lower Bound and Missing Deflation



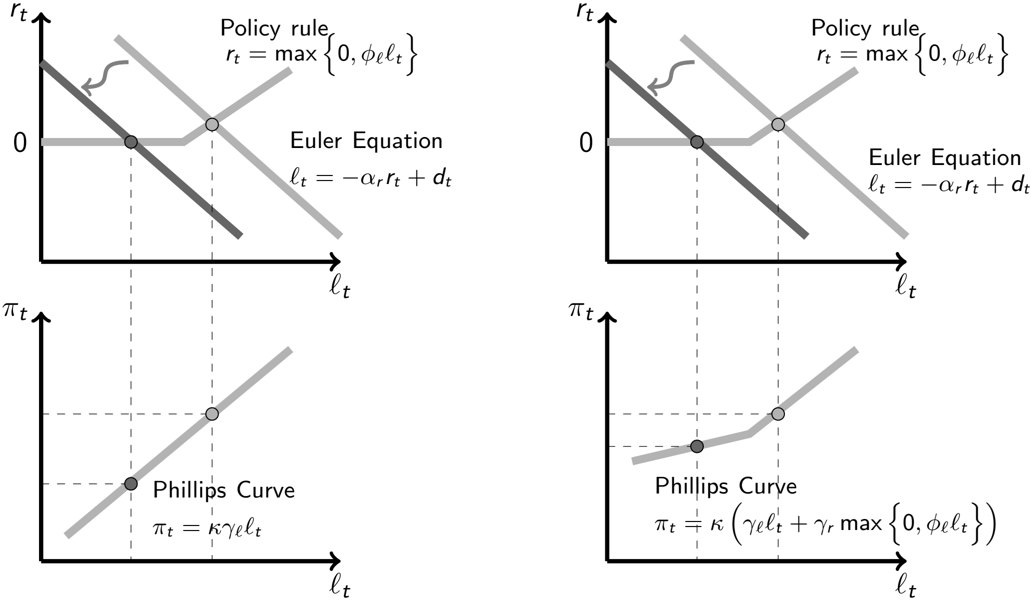
Zero Lower Bound and Missing Deflation



Zero Lower Bound and Missing Deflation



Zero Lower Bound and Missing Deflation

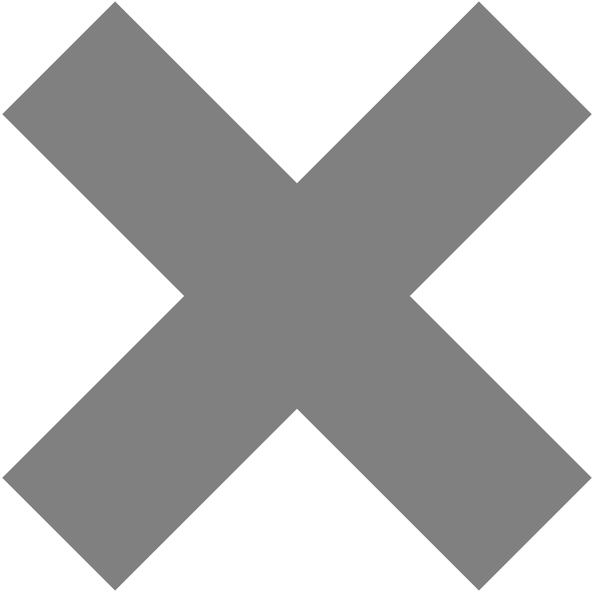


The ZLB Trap

- RK framework suggest that ZLB was quasi inevitable following a persistent fall in demand.
- In RK, both the fall in demand and the response of monetary authorities favours lower inflation:
  - Initial negative demand shock $\rightsquigarrow$
  - Low activity and low inflation $\rightsquigarrow$
  - Monetary expansion stimulus $\rightsquigarrow$
  - Lower i and lower inflation $\rightsquigarrow$
  - More monetary expansion $\rightsquigarrow$
  - Even lower i and inflation $\rightsquigarrow$
  - Hit the zero lower bound.

Summary

- When demand matters with flexible prices (*Real Keynesian* models), adding sticky prices affect the way we think of monetary policy:
  - × trade-off between stabilising inflation and output when facing demand shocks
  - × Determinacy at the ZLB
  - × Variance of inflation and output moving in opposite direction at the ZLB
- Data favours Phillips Curve with cost channel
- Data favours *Real Keynesian* configuration
- Main reason is that monetary shocks are persistent and they have neo-Fisherian effect

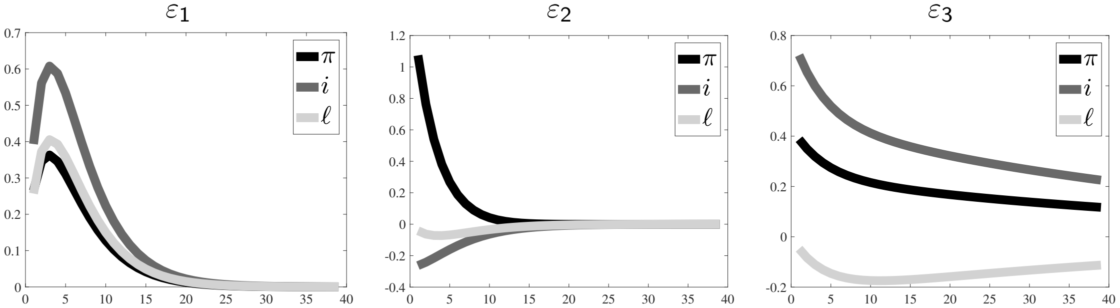


Introducing more endogenous dynamics

- Let us think of richer dynamics
  - Habit persistence
  - Hybrid New Phillips curve
  - Gradual adjustment of i
- It amounts to constraining more or less columns of A to be zero.

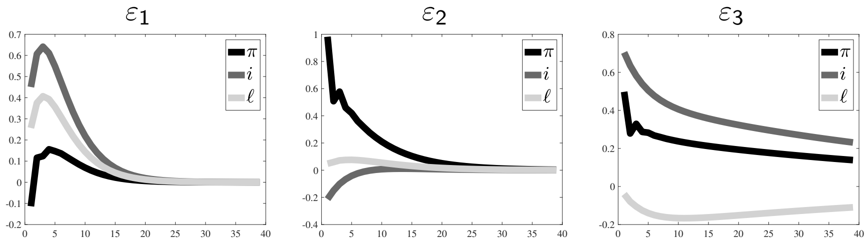
$$
\begin{aligned}
Y_t &= AY_{t-1} + BS_t \\
S_t &= RS_{t-1} + \varepsilon_t
\end{aligned}
$$

Full Sample, "Habit Persistence"

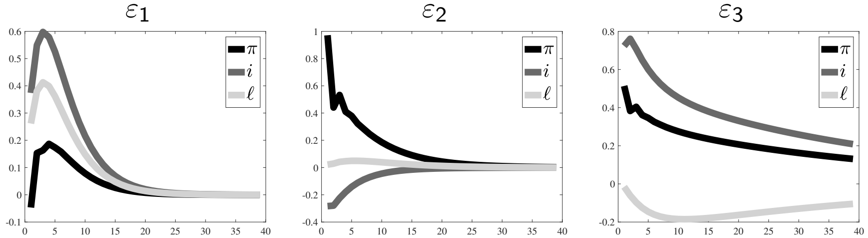


Other configurations, Full sample

“Habit persistence, and hybrid New Phillips curve”



“Habit persistence, gradual adjustment of *i* and hybrid New Phillips curve”

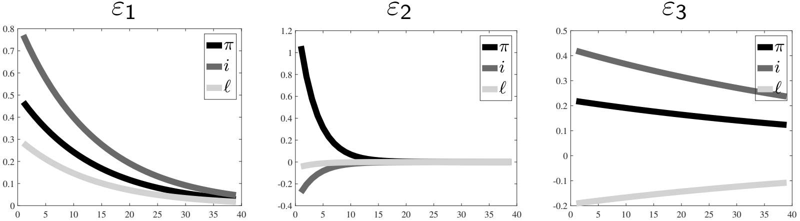


Allowing for more shocks

- Enrich the analysis by:
  - Allowing for explicit oil shocks
  - Allowing for TFP shocks
  - Allowing for natural rate of employment shocks
- We find very consistent results

Real growth (Δy) as the Fourth Variable

"Fully Forward", Full sample



"Habit persistence", Full sample